Wide Herald

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COVER STORY

Measuring the Performance of our OpenStack Cloud

Here at Target, we run our own private OpenStack cloud and have never been able to accurately measure the performance of our hardware.

Target Brands, Inc · Target Tech

This lack of measurement prevents the evaluation of performance improvements of new hardware or alternative technologies running as drivers inside OpenStack. It also prevents us from providing a Service Level Agreement (SLA) to our customers. Recently we have been striving to improve our OpenStack service which led us to talk to our consumers directly.

One of the major feedback points provided by talking with our consumers was the performance of the OpenStack cloud was lower than expected. Because we have not measured the performance of our cloud in the past, we have been unable to know if new hardware or configuration changes improves consumer-facing performance. With our new OpenStack environment builds we focused on changing this. But first we needed a tool to do the job.

id=“searching-for-a-tool”>Searching for a Tool

The first tool we looked at was Rally . Rally does performance testing of an OpenStack cloud. However, Rally focuses on the OpenStack API only. It is mainly used to test functionality (via Tempest) and stability of the API under large amounts of load. Rally does contain a resource to boot an instance and run Linux CLI commands via user data. This was tested as a way to provide the staging of instances to run performance software. However, starting the software on each instance at the same time and collecting the results from said software was difficult and not viable. Because of this, we deemed Rally was not suitable for our needs.

The next tool we looked at was KloudBuster . KloudBuster is a tool that does performance testing inside an OpenStack instance. At the time of writing it provides two sets of tests: HTTP and storage. The HTTP test uses a traffic generator to measure requests per second and latency between instances. The storage test uses FIO to measure read/write IOPs and bandwidth. KloudBuster does what we were looking for, measuring the performance of instances inside of OpenStack. However, it does not support adding more tests past the two included, it has limited configuration options for the environment setup, and it had stability issues inside our OpenStack cloud. Because of this, we deemed KloudBuster was not suitable.

id=“creating-our-own”>Creating Our Own

With current options not meeting all of our needs, we decided the best option was to create our own performance framework that can be flexible to cover a wide variety of tests and environment setups. These were our requirements:

support the following tests: network, storage, and cpu

easily support adding future tests

flexible enough to support any OpenStack environment (including old versions)

ability to use fixed or floating IP addresses for connectivity

ability to use ephemeral or cinder storage

return test results in a format usable by other software

id=“enter-cloudpunch”>Enter CloudPunch

CloudPunch is a tool developed by the OpenStack team here at Target. It is completely open source and follows the MIT license. CloudPunch has the following features:

Written 100% in Python - CloudPunch is written in the Python language including the sections that stage OpenStack and the tests that run. This was chosen to avoid reliance on other tools.

Create custom tests - Because tests are written in Python, custom written tests can be ran by simply dropping a file in a folder. These tests are not limited; a test can do anything Python can do.

Fully scalable - A test can include one instance or hundreds. A couple lines of configuration can drastically change the stress put on OpenStack hardware.

Test across OpenStack environments - Have multiple OpenStack environments or regions? Run tests across them to see performance metrics when they interact.

Run tests in an order or all at once - See single metric results such as network throughput or see how a high network throughput can affect network latency.

JSON and YAML support - Use a mix of JSON or YAML for both configuration and results

id="how-does-cloudpunch-work">How Does CloudPunch Work?

CloudPunch separates the process of running a test into three major roles:

Local Machine - The machine starting the test(s) and receiving the results outside of OpenStack.

Master - The OpenStack instance that is the communication between external and internal OpenStack. The local machine will send configuration to the master so the slaves can get it. The slaves send test results back the master so the local machine can receive them.

Slave - The OpenStack instance that runs the test(s). It reports only to the master.

For more specific information on these roles, see here

To better explain the process from start to finish, I will go over an example of running a simple ping test between instances. To initially start CloudPunch, I give the CLI a configuration file, an environment file, and an OpenRC file.

CloudPunch on the local machine then begins to stage the OpenStack cloud (in this order) with a security group, a keypair, the master router, the master network, the master instance, the slave routers, the slave networks, then finally the slave instances. The local machine now waits for all of the instances to be ready by checking in with the master server via a Flask API. At the same time, all slave instances are working to register with the master instance to provide host information and say that they are ready.

Once the master server and all slaves are registered and ready, the local machine sends the configuration to the master server and signals that the test can now begin. The slave instances are checking in with the master server every 1 second for the test status to be ready. Once it is ready, the slaves pull down the configuration from the master server. The slaves then start the ping test by calling the ping.py file inside the slave directory. This file runs the ping command via the shell and captures the latency results. The slaves collect these results and once the test is complete, send these results back to the master instance. All while this is happening, the local machine is checking in with the master server every 5 seconds for all slaves to have posted results to the master.

Once all slaves have posted results, the local machine pulls down the results and saves the results depending the configuration. Now that the process of running the test is complete, the local machine now deletes all of the resources it created on the OpenStack cloud. I am now left with the results of the ping test without having any resources sitting on the OpenStack cloud.

id="how-do-i-get-started-with-cloudpunch">How Do I Get Started With CloudPunch?

See the getting started guide here

id="about-the-author">About the Author

Jacob Lube is an Engineer and part of Target's Enterprise Cloud Engineering team focusing on anything OpenStack. Outside of OpenStack, Jacob enjoys riding bicycle, playing video games, and messing around in the Python language.

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FEATURES

A Retrospective—Responsive Design

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Not too long ago, there was a great debate between two sides of the web. The “Responsive” web vs the “m.” web. One pushing for a single codebase that responds to user-context and another pushing for a second, mobile-only approach (typically using a subdomain starting with m.) It’s clear now, the responsive web has won .

id=“a-retrospectiveresponsive-design”>A Retrospective — Responsive Design

The mark of a master woodworker is his/her ability to work with the grain. The grain gives the wood its natural identity and subtly tells the craftsman the best way to work it.

Work with the grain and save effort. Work with the grain and maximize flexibility. Work with the grain and maintain strength.

Work against the grain at the expense of additional effort on the part of the craftsman as well as reduced flexibility and strength in the wood.

The same can be said for crafting products on the web. The raw materials are HTTP requests, HTML, text, CSS, Javascript, images, and motion.

To go with the flow of the web-grain is to deliver content quickly, with vertically-variable content-flow. Reduce HTTP requests and add complex functionality using progressive enhancement.

The tools that we design should be user-centric and our metric of success is when our user puts our tools down. In the end, we want our users to be engaged with the people and world around them, not fumbling with a tool. The tools are a means, not an end.

id=“stechnmethodology”>s/Techn/Method/ology

Server-side web frameworks like Django, Ruby on Rails, Express all have excellent technologies to define explicit dependencies (e.g., pip with requirements.txt , gem with Gemfile , and npm with package.json .) Configuration management tools like Chef, Puppet, Ansible and Docker are making the server-environment both easier to define implicit dependencies and help automate the task of creating homogeneous development and production environments. Together, these tools can build a reproducible execution environment where the code can be more fully tested in a space with less far less entropy than the web environment.

The execution environment of the web is extremely heterogeneous: it operates on a fractured multitude of devices within a dozen of browser of various capability. Bandwidth, CPU speed & memory, screen size & pixel density are just a few of the multitude of variables web developers need to manage. This diversity of environment makes for some very complex problem-solving, but that very heterogeneity is the grain of the web, it’s what gives it resilience and staying-power; it’s the power of decentralization at work.

Responsive design was born out of the complexity and diversity of the web. It is not a technology but a methodology used to cope with the diversity of environments a web product needs to operate within.

We use this methodology to engage in conversation that is optimized for each of our users. We do this by taking device-capability, user attention-levels and context into account; and designing specifically for these variables. This is opposed to the familiar approach where we send a canned “one-for-all” solution to everyone (and yet no-one in particular.)

“It’s important to note the different between support and optimization. Optimizing for every web-enabled device on earth is impossible, so eventually you need to make the strategic (read: business) decision of what target devices and browsers for which to optimize.” — Brad Frost

If we bifurcate our web experiences into arbitrary “web” and “mobile” buckets, we will then be forced to decide which bucket any new device best fits in. Can we really continue to commit to supporting each new device (as the market continues to fracture) with a bespoke experience? Doubtful.

Instead we should shift our thinking: How can our designs best-adapt and flow into the context and environment in which our users operate; in order to best serve the goals of the user?

“The control which designers know in the print medium, and often desire in the web medium, is simply a function of the limitation of the printed page. We should embrace the fact that the web doesn’t have the same constraints, and design for this flexibility. But first, we must ‘accept the ebb and flow of things’.” -John Allsopp, “ A Dao of Web Design ”

Up front we should make an important distinction between “responsive design” and “web design”; namely that experience design is about purpose, while web design is about graphic aesthetics. (Most often, “web design” is just a reference to a picture, a design comp.)

While the design comp can refer to the aesthetics of the product, it doesn’t communicate its essences of motion, interaction, performance or scalability. Often times this communication exists completely outside the comp (in a discussion or meeting), or alongside it. The evidence of this gap in communication is the reliance on design-document annotations using the following words:

“Fades”

“Jumps”

“Flips”

“Shrinks”

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Installing mysql2 Gem on OSX when Headers Can’t Be Found

Scott Johnson (fuzzyblog)

Note : If you’re not a Rails user, on OSX, then just nope the heck out of reading this post. This is some serious nerdery here.

One of the single most frustrating things about the Rails ecosystem is when you can’t get a bundle install operation to finish. The bundle install, of course, is the process of getting the software tools, “gems”, identified in Gemfile to install. When you can’t get bundle install to finish, you are effectively entirely blocked from continuing with a project. A friend hit a serious problem recently with bundle install and the mysql2 gem. I gave some online help via chat and while they solved it without me, I volunteered to write it down so there’s a decent explanation / write up.

Here’s the error:

My first suggestion was to try a “brew install mariadb” (mariadb is a fully compatible, better version of MySQL, written by the creator of MySQL). It turned out that mariadb was already installed.

So this led me to this blink and then I asked for “paste me the whole disgustingly long error message on bundle install”.

current directory:

/Users/foo/.rbenv/versions/2.6.3/lib/ruby/gems/2.6.0/gems/mysql2-0.5.2/ext/mysql2

/Users/foo/.rbenv/versions/2.6.3/bin/ruby -I

/Users/foo/.rbenv/versions/2.6.3/lib/ruby/2.6.0 -r

./siteconf20191120-7300-1a2t7xt.rb extconf.rb

checking for rb_absint_size()... yes

checking for rb_absint_singlebit_p()... yes

checking for rb_wait_for_single_fd()... yes

—–

Using mysql_config at /usr/local/bin/mysql_config

—–

checking for mysql.h... yes

checking for errmsg.h... yes

checking for SSL_MODE_DISABLED in mysql.h... no

checking for MYSQL_OPT_SSL_ENFORCE in mysql.h... yes

checking for MYSQL.net.vio in mysql.h... no

checking for MYSQL.net.pvio in mysql.h... yes

checking for MYSQL_ENABLE_CLEARTEXT_PLUGIN in mysql.h... yes

checking for SERVER_QUERY_NO_GOOD_INDEX_USED in mysql.h... yes

checking for SERVER_QUERY_NO_INDEX_USED in mysql.h... yes

checking for SERVER_QUERY_WAS_SLOW in mysql.h... yes

checking for MYSQL_OPTION_MULTI_STATEMENTS_ON in mysql.h... yes

checking for MYSQL_OPTION_MULTI_STATEMENTS_OFF in mysql.h... yes

checking for my_bool in mysql.h... yes

—–

Don’t know how to set rpath on your system, if MySQL libraries are not in path

mysql2 may not load

—–

—–

Setting libpath to /usr/local/Cellar/mariadb/10.4.10_1/lib

—–

creating Makefile

current directory: /Users/foo/.rbenv/versions/2.6.3/lib/ruby/gems/2.6.0/gems/mysql2-0.5.2/ext/mysql2 make “DESTDIR=” clean

current directory: /Users/foo/.rbenv/versions/2.6.3/lib/ruby/gems/2.6.0/gems/mysql2-0.5.2/ext/mysql2 make “DESTDIR=” compiling client.c client.c:140:1: warning: control may reach end of non-void function [-Wreturn-type] } ^ 1 warning generated. compiling infile.c compiling mysql2_ext.c compiling result.c compiling statement.c linking shared-object mysql2/mysql2.bundle ld: library not found for -lssl clang: error: linker command failed with exit code 1 (use -v to see invocation) make: *** [mysql2.bundle] Error 1

make failed, exit code 2

Gem files will remain installed in /Users/foo/.rbenv/versions/2.6.3/lib/ruby/gems/2.6.0/gems/mysql2-0.5.2 for inspection. Results logged to /Users/foo/.rbenv/versions/2.6.3/lib/ruby/gems/2.6.0/extensions/x86_64-darwin-18/2.6.0/mysql2-0.5.2/gem_make.out

An error occurred while installing mysql2 (0.5.2), and Bundler cannot continue. Make sure that gem install mysql2 -v ‘0.5.2’ –source ‘https://rubygems.org/’ succeeds before

yes checking for SERVER_QUERY_WAS_SLOW in mysql.

Scott Johnson (fuzzyblog)

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Java 9 modules - JPMS basics

Stephen Colebourne · Stephen Colebourne (Joda)

The Java Platform Module System (JPMS) is the major new feature of Java SE 9. In this article, I will introduce it, leaving most of my opinions to a follow up article. This is based on these slides .

Java Platform Module System (JPMS)

The new module system, developed as Project Jigsaw , is intended to raise the abstraction level of coding in Java as follows:

* Make the Java SE Platform, and the JDK, more easily scalable down to small computing devices;

* Improve the security and maintainability of Java SE Platform Implementations in general, and the JDK in particular;

* Enable improved application performance; and

* Make it easier for developers to construct and maintain libraries and large applications, for both the Java SE and EE Platforms.

To achieve these goals we propose to design and implement a standard module system for the Java SE Platform and to apply that system to the Platform itself, and to the JDK. The module system should be powerful enough to modularize the JDK and other large legacy code bases, yet still be approachable by all developers.

However as we shall see, project goals are not always met.

What is a JPMS Module?

JPMS is a change to the Java libraries, language and runtime. This means that it affects the whole stack that developers code with day-to-day, and as such JPMS could have a big impact. For compatibility reasons, most existing code can ignore JPMS in Java SE 9, something that may prove to be very useful.

The key conceptual point to grasp is that JPMS adds new a concept to the JVM - modules. Where previously, code was organized into fields, methods, classes, interfaces and packages, with Java SE 9 there is a new structural element - modules.

a class is a container of fields and methods

a package is a container of classes and interfaces

a module is a container of packages

Because this is a new JVM element, it means the runtime can apply strong access control. With Java 8, a developer can express that the methods of a class cannot be seen by other classes by declaring them private. With Java 9, a developer can express that a package cannot be seen by other modules - ie. a package can be hidden within a module.

Being able to hide packages should in theory be a great benefit for application design. No longer should there be a need for a package to be named “impl” or “internal” with Javadoc declaring “please don’t use types from this package”. Unfortunately, life won’t be quite that simple.

Creating a module is relatively simple however. A module is typically just a jar file that has a module-info.class file at the root - known as a modular jar file . And that file is created from a module-info.java file in your sourcebase (see below for more details).

Using a modular jar file involves adding the jar file to the modulepath instead of the classpath. If a modular jar file is on the classpath, it will not act as a module at all, and the module-info.class will be ignored. As such, while a modular jar file on the modulepath will have hidden packages enforced by the JVM, a modular jar file on the classpath will not have hidden packages at all.

Other module systems

Java has historically had other module systems, most notably OSGi and JBoss Modules. It is important to understand that JPMS has little resemblance to those systems.

Both OSGi and JBoss Modules have to exist without direct support from the JVM, yet still provide some additional support for modules. This is achieved by launching each module in its own class loader, a technique that gets the job done, yet is not without its own issues.

Unsurprisingly, given these are existing module systems, experts from those groups have been included in the formal Expert Group developing JPMS. However, this relationship has not been harmonious. Fundamentally, the JPMS authors (Oracle) have set out to build a JVM extension that can be used for something that can be described as modules, whereas the existing module systems derive experience and value from real use cases and tricky edge cases in big applications that exist today.

When reading about modules, it is important to consider whether the authors of the article you are reading are from the OSGi/JBoss Modules design camp. (I have never actively used OSGi or JBoss Modules, although I have used Eclipse and other tools that use OSGi internally.)

module-info.java

The module-info.java file contains the instructions that define a module (the most important ones are covered here, but there are more). This is a .java file, however the syntax is nothing like any .java file you’ve seen before.

There are two key questions that you have to answer to create the file - what does this module depend on, and what does it export:

```
}
{
```

Meet Tuhina, Software Engineering Intern @ BigCommerce

Tuhina Das · BigCommerce Engineering

Hi! I'm Tuhina Das, and I'm a rising senior in high school. This summer, I interned with the Mobile Team here at BigCommerce.

Ever since taking my first coding course at the start of high school, I've always had a profound interest in building software. So naturally, I wanted to learn more about different specialties in software engineering and the tools they involve. This ultimately led me to take courses with Code2College (C2C) in 2023, a nonprofit striving to "increase the proportion of historically [underrepresented] high school students who enter and excel in STEM… careers." That fall, I joined C2C's Internships program, which partners with companies to place students in technical internships. Through this program, I got to meet the incredible Mobile Team for the first time.

Going into this internship, I was very excited because I finally could get a feel for what it's like being a software engineer, a career path that I was seriously considering. I was also looking forward to meeting and getting to know people at the company, both in and out of the team.

My first two weeks at BigCommerce was a sort of 'set-up' period – I, along with my fellow intern Arushi, completed our onboarding tasks and kept track of different tools we could explore the next few weeks through a 30, 60, and 90 day plan that the team provided. After completing all our required training and receiving access to the team's codebases, we were finally assigned our first ticket: to implement stylized designs of an in-app satisfaction prompt for the BigCommerce iOS app. The prompt included a header, description, and buttons to navigate through it and close it.

Using Swift, the coding language used to power iOS apps, for the first time and navigating the expansive codebase that the team had put together over the course of years was somewhat daunting. But the Mobile Team was more than happy to help me learn the ropes of building my first Swift project. By the end of my third week, I had started creating a mock-up of the prompt's user interface.

After building the mock-up, I started working on implementing it in the Mobile Team's codebase. Arushi and I met daily with Tina Ho and Kymarly Henry , two software engineers on the team who mentored us throughout our internship, to better understand how to build and style the prompt with SwiftUI. And before we knew it, we had our first pull requests (a request to merge your code into the codebase) for our assigned tickets, and the iOS prompt was finished!

We then started working on the prompt for the Android app, which was starting to feel less daunting now that we had created one for the iOS app. We spent the next two weeks familiarizing ourselves with the Android codebase bit by bit, and then started implementing the second prompt. We are just beginning to wrap up this prompt, and hopefully we'll see it released sometime in the near future.

I have a lot of fond memories of my internship this summer, so I'd like to highlight a couple with you that I think best reflect my experiences as a BigCommerce intern.

I'll start off with Friday Throwdown meetings. Every Friday afternoon, the Engineering Department at BigCommerce meets up to catch up with each other. The agenda covers a wide variety of topics, from Big Wins to anniversaries/birthdays/pets/new hires to executive announcements and dad jokes. My first week, I was admittedly a little nervous to introduce myself to such a large audience. But in the weeks after, I felt as if I were right at home; there's an evidently warm and collaborative environment fostered by everyone in the department.

I also had the pleasure of meeting Brent Bellm and Travis Hess, the CEO and President of BigCommerce , respectively. As a teenager constantly seeking guidance on how to navigate adulthood, I thoroughly enjoyed getting to hear about their experiences and lessons they learned in life. Amidst our conversation, we discussed the importance of investing in taking care of yourself and how that could be a powerful driving force for motivation. We also spoke about not shying away from exploring different careers of interest. It was inspiring to speak with Brent and Travis and get to know them, and I'm incredibly thankful to be able to have done so.

BigCommerce's commitment to establishing a sense of community amongst its employees – as well as its emphasis on taking care of them – was evident in the past eight weeks I spent interning here. It's why my experience has been a memorable one.

I learned a significant amount about a software engineer's day-to-day activities, and I can confidently say that I've grown not only professionally, but also personally because of this internship. I'd be in a very different place today without having worked here, and without the Mobile Team's guidance and support. Because of them, I know that this is a field I definitely want to pursue.

Special thanks to the Mobile Team, and thank you, BigCommerce! Think Big!

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Should you adopt Java 12 or stick on Java 11?

Stephen Colebourne · Stephen Colebourne (Joda)

Should you adopt Java 12 or stick on Java 11 for the next 3 years? Seems like an innocuous question, but it is one of the most important decisions out there for those running on the JVM. I'll try to cover the key aspects of the decision, with the assumption that you care about running with the latest set of security patches in production.

TL;DR, It is vital to fully understand and accept the risks before adopting Java 12.

The Java release train

There is now a new release of Java every six months, so Java 12 is less than five months away despite Java 11 having just been released. As part of the process of moving to more frequent releases, certain releases are designated to be LTS (long term support) and as such will have security patches available for four years or more. This makes them "major" releases, not because they have a bigger feature set but because they have multi year support.

It is expected that Java 11 patches (11.0.1, 11.0.2, 11.0.3, etc.) will be smaller and simpler than Java 8 updates (8u20, 8u40, 8u60, etc.) - Java 11 updates will be more focused on security patches, without the internal enhancements of Java 8 updates. Instead, Oracle want us to think of Java 12, 13, 14 etc. as small upgrades, similar to an imaginary Java 11u20, 11u40 etc. To be blunt, I find this nonsensical.

Senior Oracle employees have repeatedly argued that updates such as 8u20 and 8u40 often broke code. This was not my experience. In fact my experience was that update releases primarily contained bug fixes. The only break I can remember was the addition of –allow-script-in-comments to Javadoc, which isn't a core part of Java. As a result, I have never feared picking up the latest update release - and this has been a core benefit of the Java platform.

Drilling down into why update releases tend to cause no problems, lets examine the differences between release types:

Model Old model New model

Upgrade Java major releases Java update releases Java release train Java patches

Frequency Every 3 years or so Every 6 months Every 6 months Every 3 months

Versions 6 -> 7 -> 8 8 -> 8u20 -> 8u40 11 -> 12 -> 13 11 -> 11.0.1 -> 11.0.2

Language changes ⍰ ✘ ⍰ ✘

JVM changes ⍰ ✘ ⍰ ✘

Major enhancements ⍰ ✘ ⍰ ✘

Added classes/methods ⍰ ✘ ⍰ ✘

Removed classes/methods ✘ ✘ ⍰ ✘

New deprecations ⍰ ✘ ⍰ ✘

Internal enhancements ⍰ ⍰ ⍰ ✘

JDK tool changes ⍰ ⍰ ⍰ ✘

Bug fixes ⍰ ⍰ ⍰ ⍰

Security patches ⍰ ⍰ ⍰ ⍰

Given the table above, I find it amazing that anyone would claim moving from 11 to 12 to 13 is anything like moving from 8u20 to 8u40. But that is the official Oracle viewpoint:

Oracle FAQ

As the table clearly shows, each version in the Java release train can contain any change traditionally associated with a full major version. These include language changes and JVM changes, both of which have major impacts on IDEs, bytecode libraries and frameworks. In addition, there will not only be additional APIs, but also API removals (something that did not happen prior to 8).

Oracle's claim is that because each release is only 6 months after the previous one, there won't be as much "stuff" in it, thus it won't be as hard to upgrade. While true, it is also irrelevant. What matters is whether the upgrade has the potential to damage your code stack or not. And clearly, going from 11 -> 12 -> 13 has much greater potential for damage than 8 -> 8u20 -> 8u40 ever did .

The key difference compared to updates like 8u20 -> 8u40 are the changes to the bytecode version, and the changes to the specification. Changes to the bytecode version tend to be particularly disruptive, with most frameworks making heavy use of libraries like ASM or ByteBuddy that are intimately linked to each bytecode version. And moving from 8u20 -> 8u40 still had the same Java SE specification, with all the same classes and methods, something that cannot be relied on moving from 12 to 13. I simply do not accept Oracle's argument that the "amount of stuff" in a release is more significant than the "type of stuff".

Note however that another one of Oracle's claims really does matter. They point out that if you stick with Java 11 and plan to move to the next LTS version when it is released (ie. Java 17) that you might find your code doesn't compile. Remember that the Java development rules now state that an API method can be deprecated in one version and removed in the next one. Rules that do not take LTS releases into account. So, a method could be deprecated in 13 and removed in 15. Someone upgrading from 11 to 17 would simply find a deleted API having having never seen the deprecation. Lets not panic too much about removal though - the only APIs likely to be removed are specialist ones, not those in widespread use by application code.

Considerations before adopting the release train

In this section, I try to outline some of the considerations/

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Equity Allocation in Startups

Scott Johnson (fuzzyblog)

Once upon a time, I was speaking with a company founder and they mentioned that they had a VP of Engineering to whom they have given a 1/3 stake in the company. I immediately commented that was too much and then said “I’ll write this down in a blog post” – and then I never did. Last night, oddly, I woke up from a deep sleep with the desire to write this down. And that brings us to this post.

Here’s what I can remember from that conversation:

Company Stage : Pre Funding

Company Type : Medical

Equity Split with the VP of Engineering : 1/3

Founder Title : Yes

Did VP of Engineering Put in Cash : No

id=“you-need-to-understand-this”>You Need to Understand This

One of the basic rules of the startup world is that on Day 1, you, the founder, own 100% of something that is worth absolutely nothing . The goal, by the end, is that you own a much smaller percentage of something actually worth something. As an example, owning 10% of something worth $10 million is actually much, much better.

id=“the-basic-rules-of-thumb-for-equity-allocations”>The Basic Rules of Thumb for Equity Allocations

Here are my rules of thumb to use for equity allocation:

The more risk you take, the more you get

The earlier you join, the more you get

Putting time in is one type of risk

Putting cash in is a greater type of risk

If you, the founder, give too much equity to someone else then you can be pushed out by simply having that other person align with the investor or investors

You only have 80% of the equity to play with - 20% generally goes to an ESOP (employee stock option plan)

Make damn sure that you give out options not stock and a long vesting schedule (incremental over say 4 years)

The bottom line is that equity, in whatever form, is a reward for taking risk. And the earlier you are involved in a startup, the more risk there is.

id=“my-personal-experience-from-feedster”>My Personal Experience from Feedster

A long, long time ago, I founded a blog search engine named Feedster. I merged with another RSS search engine shortly after coming to market to address some technical limitations in my architecture. We did the typical nerd founder thing and simply split the equity down the middle and both took the Founder title.

In retrospect that wasn’t a fair allocation because I committed significantly more time to Feedster but it avoided a difficult conversation that I was simply not brave enough to have – and that was a mistake.

Once we realized that blog search was actually a viable business, my first hire was a CEO to handle the operations and capital raise. This CEO:

Came in after the technology was built

Came in after the site had traffic

Came in after growth was proven

Did not have to put in any personal money

The equity allocation for that CEO, bear in mind that this was 16 years ago so my memory may be fuzzy was between 9% and 10%. That CEO’s advice to me is that if we filled other VP level slots, we’d be talking about between a 3% to 5% stake.

If you contrast these numbers with the 1/3 equity split mentioned above, it is considerably less and the reason was that an awful lot of the risk had already been addressed (functional technology, working product, site with traffic, no capital investment, etc).

id=“disclaimers”>Disclaimers

There are two important disclaimers to understand here:

I am not a lawyer (IANAL) and my focus is always on making great technology NOT on the best personal economic return. This means that there are aspects of capital raising and equity allocation that I don’t now and likely never will fully understand.

I should also note that equity allocation is a particularly sensitive topic for me because, even though I was the founder, I was forced out by the people I hired and then voted off the board. And while I am over it (somewhat), had the equity allocations been handled differently, that might not have happened.

id=“what-should-this-founder-do”>What Should This Founder Do?

My advice to this founder is to get Marc Randolph’s book, That Will Never Work , about the founding of Netflix and read pages 180 to 189. Marc Randolph was the founder of Netflix and the original CEO. In these pages, Reed Hastings not only told Marc that he needed to step down as CEO and be replaced by Reed but that he needed to give a substantial

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I did an inktober and I want to tell you about it

Monica Dinculescu

Inktober is a project where artists make an ink drawing every day for the whole month of October. This year I did an inktober but ignored all the rules, and made Internet Stuff™ instead. That experiment lives here , but I want to tell you why I did it before you go ahead and judge it. I think that it's also important to tell you it was a huge pain in the ass just in case you watched it unfold and thought I magically stuck every landing.

id="the-why">The why

"Art" is a word I struggle with a lot. I don't think of myself as an artist, because none of the things I make feel like art yet. At the same time, I obsess about these things if I don't feel they represent me , which is one of the nigthmares of being an artist – hence the struggle. For example, I made pottery for well over a year before I thought any of the mugs looked like what I wanted them to look like. They looked fine, and they looked like mugs, but they weren't the kind of mugs I wanted to make.

In my opinion, one of the qualities you need to be an artist is to be extremely creative. I am not extremely creative, but I believe that everything in life can be learnt by grinding that level. So with inktober I wanted to force myself to think creatively every day, for a whole month, and see if that would level up my creativity.

id="the-how">The how

I don't like open ended projects because I think they lead to intellectual wankery instead of like, actually doing the thing , so I made up these rules for my weird art experiment:

it had to be somehow about the sheriff of circle punch [1]

it shouldn't take more than like, an hour

it's fine if I skipped weekends [2]

[1] Honestly the first day of inktober was a joke to make fun of the hashtag and my friend Fabian , who I troll with a circle punch any time I can. This of course ended up trolling me, because I did the whole fucking month afterwards, so ain't that the troller becoming the trollee. My friend Bushra came up with the name ("it's always howdy, never howthee"). Her inktober was way better.

[2] I don't work weekends, and if anyone tells you that "being creative", or "making art" doesn't feel like work, don't believe them. Nothing is a free lunch.

id="the-aftermath">The aftermath

I learnt a lot of CSS. A lot of the days ended up with me looking at Codepen and trying to figure out what animation I could use that day. I learnt about svg filters, and became really comfortable writing keyframes. I finally managed to write align-items: center; justify-content: space-between; without having to look it up. I only crashed my browser once.

Overall it was really, really hard, and I hated a lot of it. A bunch of the days were political . Some of the days were freebies . Day 18 was my favourite. Day 10 almost made me throw my laptop at the cat.

And you know what? I think it worked. I noticed that I was walking around all day looking at everything, like artists look at everything, trying to see what I could use for that day's inktober. Once, when I was in highschool, an art teacher set me up on a mentoring date with an art school major. I told her half of the time I didn't know what to paint, and she told me to just look at things and make them weird. "It's like the Eiffel tower, but like melting over a bridge, you know?". I didn't know, but then on day 18 I took a drum beat and made it into a Franken-morphed sheriff. I wrote down every idea that I had, and after the month was over, I still had ideas left over. They're not all good, but you can afford being picky when you're out of the drought.

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REDstack

Target Brands, Inc · Target Tech

id=“redstack-is-now-open-source”>REDstack is Now Open Source!

We are officially open sourcing REDstack, our sandbox tool for Big Data development at Target.

id=“what-is-redstack”>What is REDstack?

REDstack is a tool for provisioning kerberized clusters on OpenStack. We created it with four goals in mind:

Provide a secured environment, with the ability to leverage preconfigured LDAP and Kerberos servers.

Out of the box usability, allowing you to log in with preconfigured user accounts.

Custom user management utilities to administer the cluster.

Provide a fully customizable experience, everything is a configuration option in your build files:

Cluster size, node sizes, types of nodes and node roles,

Hadoop configurations, heap sizes, and components,

All users, passwords, and secure assets.

id=“components”>Components

REDstack is made up of two major components:

hdp-cloud - The cookbook

The cookbook is used by the application itself to install components and lay down cluster configuration.

The cookbook can be used independently of REDstack to manually provision a cluster.

REDstack - The orchestration component

> *id=“a-product-is-born”>A Product is Born By this point, we had written the entire process as a Python application and set it up on a nightly loop.*
>
> Target Brands, Inc

REDstack is a python application that performs all of the high-level complexities and timings associated with a full Hadoop installation:

Orchestrates the provisioning of resources over OpenStack APIs,

Controls and monitors parallel Chef deployment across the cluster,

Manages and monitors cluster component install over HTTPS requests.

REDstack is bundled with a Docker image, where the configs are set up locally before an installation, and all of the dependencies are updated and configured.

id=“how-to-get-started”>How to Get Started

Head over to the repository at https://github.com/target/redstack and follow along. The repo has instructions on how to build and configure the clusters using the included Docker image.

id=“history-of-the-project”>History of the Project

Target’s Big Data Platform Team manages multiple Big Data environments, with hundreds of nodes and many PB’s of data. As mentioned in our prior blog posts, we depend heavily on Chef as a core part of our CI/CD pipeline. During our testing a couple of years ago, we identified a large opportunity to provide a way to do full integration testing with our Chef cookbooks. This opportunity opened the door for a new product.

id=“origins”>Origins

Early on, we released a product internally called Pushbutton. This was a three-node cluster that ran on a standard issue laptop at Target. It was kerberized, and it worked well as a Sandbox environment for developers because it had the same security setup. PushButton, however, did not use the same cookbooks as the main cluster. We wanted something that could do what PushButton did, but with our real cookbooks in a larger environment. By creating a full cluster from scratch, we would be able to understand exactly how the cookbooks would function when applied to new nodes and make sure the cookbooks were in a constant working state. We also needed a little bit larger of a sandbox, otherwise we would have difficulty testing high availability (HA) components, or those that run across multiple nodes, like Apache Zookeeper.

id=“toward-redstack”>Toward REDStack

Our early exploration started out by trying to adapt the existing PushButton work onto OpenStack. We used shell scripts to automate creation of instances, Knife to bootstrap the nodes with the Chef recipes, and cURL requests to automate and monitor the install process. We got it working, but we still had to face our biggest challenge yet, integrating the cookbooks meant for physical hardware onto virtualized nodes. They were expecting particular configurations such as physical drive formatting and partitioning, or where master services are already defined and running. Instead, we had to get them working with our minimum 1x50GB virtual volumes, and anything we changed would have to still be working on the physical nodes. After some difficult

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The perils of tensor.dataSync()

Monica Dinculescu

One of the first things you stumble on when you start using TensorFlow.js is that sometimes you need your data as a Tensor, and sometimes you need it as a JavaScript number. Maybe it’s for logging it, maybe it’s for displaying it somewhere during training, maybe it’s because you don’t trust the robots to be better than you at math.

This is a quick post that tries to clarify why doing this synchronously is probably bad and will leave your UI really janky. Nikhil (who like, birthed TensorFlow.js, bless) was kind enough to explain this to me recently, so I figured I’d return the favour, with fewer meeps and more mistakes.

id=“downloading-and-uploading”>Downloading and Uploading

When you create a Tensor, it lives on the CPU. The mere fact that it’s a Tensor doesn’t automatically move that data into its GPU mansion – it needs to be used in a WebGL program. (I’m playing fast and loose here with the words GPU and CPU btw, so hold back the pedantics: when I say “it lives on the CPU” I mean “in main memory, where the CPU processes stuff”; the GPU has it‘s own memory, that’s where it processes stuff, and that’s where that data has to get transferred to. It’s fine. You know I know.)

You upload the tensor to the GPU when you call one of the tf. operations on it. Tensor operations are matrix math, and matrix math is really fast on the GPU, so every time you call something like sum or sqrt on a Tensor, TensorFlow.js creates a little WebGL operation , and sends it to the backend. Whatever data lived on the CPU is now “uploaded” to the GPU (to a WebGL texture).

You download a Tensor when you want to get that data from the GPU back onto the CPU. The data now lives in a WebGL texture, so TensorFlow.js needs to call readPixels to ... read... those pixels... from the texture and convert them back into something you can use. Here’s the problem: calling readPixels is fundamentally a blocking operation: when you ask the GPU to give you data, you have to wait for it to respond; this means you can’t really do anything else on the screen while this is happening, like paint any animations.

TL;DR:

class=“highlight”> const a = tf.tensor(); /\/ a is on the CPU. const b = a.sqrt(); /\/ Upload a’s data to the GPU. const c = a.dataSync(); /\/ Download a’s data from the GPU to the CPU.

So the problems here are:

calling readPixels will make your UI janky.

downloading and uploading from the GPU isn’t free, so doing this over and over is bad news bears.

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downloading from the GPU synchronously over and over is a 2-in-1 and will probably murder your favourite pet.

id=“how-it-works”>How it works

If you read the latest 0.15.1 docs, you’ll discover that there are at least 4 ways of “downloading” your tensor:

aTensor.array() – asynchronous, and keeps the shape of the tensor (so it returns a nested array)

aTensor.arraySync() – same as above but synchronous

aTensor.data() – asynchronous and doesn’t keep the shape of the tensor (and returns a fancy Float32Array like type)

aTensor.dataSync() – same as above but synchronous

Out of these, I personally prefer the new array flavours, since I think about my tensors based on their dimensions, so when they get flattened I get confused.

The difference between the sync and async versions is that:

for the sync methods, TensorFlow.js just goes ahead and calls readPixels , which instantly blocks and causes sadness.

for the async methods, it creates a “fence” (think of it like a fancy WebGL setTimeout), and then calls a different method, getBufferSubData when that fence is hit. Unlike readPixels , this doesn’t block the UI thread and it won’t cause sadness.

If you, like me, have strange hobbies and want to find this in the actual TensorFlow.js source code, check out the read and readSync methods in this file .

id=“what-to-do”>What to do

My advice is:

if you have to download your data, try to do it once, asynchronously. Do this at the end, after all your GPU computations are done.

reach towards the async versions first – that way, even though the operation is expensive, it won’t block the UI and you can do other non-janky things like letting the user scroll on the page.

if you really really pinky swear have to use the sync version, just take another look at the code and see if you can’t move that call somewhere else where it can by async.

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A Conceptual Architecture for a Filesystem to SQS Loader

Scott Johnson (fuzzyblog)

I have an interesting technical situation facing me:

A docker container which runs persistently and writes JSON files to a directory

Each JSON file contains short form messages with JSON encoding along a variety of metadata fields

Each file is idempotent i.e. once it exists in the directory will never be written to again

Each file is named for a timestamp

Each file needs to:

Have ridiculous, non-JSON comments removed from the beginning

Be parsed into json

There is roughly 6000 of these JSON things coming in per second

Have only a selection of the JSON internally be selected

Be batched into a collection up to the maximum size the SQS supports and then committed to SQS

Be processed only once

Note : SQS is the AWS "Simple Queue Service". A queue is a specialized data structure which hands data off for processing by other tasks.

My initial thinking to handle this was to use Rust to write a high performance file processor. This appealed to me:

New technical challenge; I don't know Rust

Manly - writing fast code is a conveniently macho challenge

Simple architecture - one box, one process watching a directory and ripping through files

There are, naturally, problems here:

The Rust program represents a SPOF or "Single Point of Failure"; things break

This architecture is simple but making it not be a SPOF means the architecture gets complex

We don't actually even know if a single Rust executable is fast enough to handle all the data

Rust is complex and even simple things take dramatically more code than a scripting language like Ruby or Python. There isn't anything wrong with Rust (I really like the language but learning it is non-trivial).

Thinking through all these issues as well as learning, by chance, that the runtime execution period for AWS lambda serverless functions had increased from 5 minutes to 15 minutes made me think in terms of a different architecture focused on using lambdas.

Note : A lambda is a self contained bit of code that you give over to AWS to manage on your behalf. Another term for Lambda is "functions as a service". You don't have to focus at all on servers, DevOps administration or the like.

Here is what I'm thinking:

Add a network API to the filesystem of JSON files. This could literally be as simple as an NGINX server that listed the files.

A lambda that requests a JSON file for processing per the description above and relies on a Redis dictionary to track files that have already been processed. Two dictionaries would be needed – json_files_processed and v (and, yes, there would need to be a way to expire things from json_files_processing in case a lambda crashes or is terminated; this would be a separate lambda).

A CloudWatch Scheduler rule that triggers the lambda every 5 minutes

Fleshing this out further gives three lambda functions:

sqs_loader - the main thing which processes files and sends their contents to SQS

json_files_processed_archiver - pulls files that have been processed into a different directory so that they don't have to be ever checked again; alternatively this might delete them after some period of time

json_files_processing_checker - checks to see if files have been in the json_files_processing queue for too long

There would likely need to be 3 CloudWatch Scheduler rules one for each lambda.

The sqs_loader would need the ability to self terminate / exit if all files are currently being processed.

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Essential Ember Addons: The State of the Ember Addon Ecosystem in 2019

0xADADA

2019 has been a great year for Ember so far, so while my peers are focused on setting direction for the framework for the rest of 2019, I wanted to take stock of the existing addons ecosystem.

In this article I'd like to present a list of Ember addons that I use in most of my projects. I've been using Ember for the last few years as my go–to framework for developing web applications, and many of these addons make appearances in nearly all of them.

Ember addons generally fall into one (or more) category of functionality I'll be referring to throughout this guide:

Build-time Build-time addons provide command-line tools that help developers during the creation of the application. An example is ember-cli-eslint which provides code linting, or ember-cli-typescript which adds a build pipeline for transforming TypeScript files into JavaScript files. These addons don't ship features to your deployed application. These addons typically start with the prefix ember-cli- .

Runtime Runtime addons provide features that will be present in the final application, these include Ember components like ember-power-select and ember-svg-jar these addons increase the payload of the deployed application. These addons typically start with the prefix ember-

Infrastructure Infrastructure addons provide features that aren't shipped with the payload of your application, but provide functionality that improves the development ergonomics or deployment of the project. For example ember-cli-fastboot provides a backend Node.js server for rendering Ember apps serverside.

Quality Assurance Quality assurance addons provide tools for improving quality of code over time, and improving the developer experience of writing and testing code. These addons typically provide functionality that is used at build and test time, but isn't shipped to your deployed application. qunit-dom , coveralls , and ember-test-selectors are examples of quality assurance addons.

Some of these addons are included by default by ember new but I'll elaborate on their use a bit more.

Finally, before diving into the addon list, I won't be discussing many standard JavaScript packages. There are a bunch of JavaScript packages that I often use (ramda , lodash , etc) but these are outside of the scope of this article.

id="contents">Contents

id="general-purpose-addons">General Purpose Addons

These addons are used in nearly all my projects, I often install and configure them right after I've created a new project.

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Cat-DNS: a DNS server that resolves everything to cats

Monica Dinculescu

The internet needs more cats. DNS servers are the authority on all things internet. Therefore, the best DNS server is the one that resolves everything to cats. Guess what kind of DNS server this is (Hint: it’s the cat kind).

id=“making-it-go”>Making it go

First, get the code , and the npm packages you need (the instructions are with the code). To run, start the server as a privileged process. This is because to be a DNS server, you need to be a UDP server on port 53. This is a small numbered port, which means it needs superpowers. This is how your run it:

sudo node cat-dns.js

You also need to somehow set your DNS server to be localhost. On a Mac, I do this by creating a new (wi-fi) interface (called Cats), in my Network preferences, and setting its DNS server to 127.0.0.1 . You could do this on your normal interface, but this makes switching back and forth easier.

id=“warnings”>Warnings

While you’re playing with this, pretty much nothing on your computer that requires the internet works. Except for your browser. And then that’s mostly cats. So being able to deactivate this easily is kind of key (I know. You might think “Why would I ever want to deactivate cats?”, but trust me on this one). I also recommend killing all the things that need to call the mothership (google hangouts, twitter feeds, dropbox, iMessage), because they will not like your sassy cat answers, and will slow everything down.

id=“you-are-ready”>You are ready

Go in your browser to www.google.com and wait a bit. You should see a cat. Go to a different website. Another cat. Congratulations. Your internet is now all cats.

id=“wait-what”>Wait, what?

Do not panic. While I recommend you don’t look at the source because it’s gross, if you do look at the source, you’ll notice all it does is resolve any hostname to 54.197.244.191 , which is a magical place on the internet that has cats. My friend @eigma made it, and is hosting it, so please try not to kill all the cat bandwith at once. You could also resolve everything to localhost, and serve your own for now cats on an http server on port 80. But then you’d have to store your own cats locally, and that is animal cruelty. Thankfully, for now, while that magical static IP exists, you don’t have to. That’s it, that’s all.

id=“i-need-to-know-more”>I NEED TO KNOW MORE

Here’s the little summary I wrote originally about how DNS servers work. Basically, cat-dns ends up doing this:

gruesomely parse the query from the client. I used this as a reference on what each of the fields in the message sections are, because the spec itself is very dry. This was the worst part, because the message sections are sequences of bits that don’t add up to bytes on any sane boundary, so you have to work with bit arrays, which is nobody’s idea of fun. Anyway, a spec is a spec.

assemble the DNS answer. The answer is mostly the same for each query – the only thing that changes is the content of the query (i.e. the hostname you requested). And you copy that from the query, so it’s not a big deal.

always returns 54.197.244.191 as the resolved IP, unless you’re requesting imgur.com . Then it gives you an actual IP for imgur that I got from nslookup . Imgur stores our cats, so we need to be able to get to them. :)

> *You could also resolve everything to localhost, and serve your own for now cats on an http server on port 80.*
>
> Monica Dinculescu

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Scott’s Approach to Weight Loss

Scott Johnson (fuzzyblog)

About two years ago, I lost 20 pounds. I compared experiences recently with a friend who had lost a bunch of weight and our experiences were so different that I thought I’d write them down. I’m also writing this down because my eldest son just enlisted in the Marines and he needs to drop weight before he goes to boot camp.

Here’s what worked for me:

Weigh Yourself Every Day .

Whole 30 .

Zero App .

Exercise At Least 3 times Per Week .

No Dining Out .

Each of these is discussed below.

id=“weigh-yourself-every-day”>Weigh Yourself Every Day

I’m an engineer, not a weight loss person, but, to my way of thinking, if you’re going to try and lose weight then you need a metric and that metric is your weight . I’ve now been weighing myself daily since 2017 and I can tell you that weighing daily really helps prevent you from gaining weight.

Note 1 : If you need a way to track your weight on a daily basis, reach out to me. I have a new product coming out that helps with this.

Note 2 : If you are going to weigh daily then you need to come up with a standard approach. For me that is:

Wake up.

Use the bathroom.

Weigh; always using the same scale.

Log my weight.

id=“whole-30”>Whole 30

I used the Whole 30 approach to diet. This basically boils down to:

Whole Foods - simple proteins and vegetables

Nothing artificial; everything made from scratch right down to mayonnaise

No sugar at all including sugar, honey, maple syrup, etc.

No grains

No alcohol

No beans / legumes

No dairy (including milk, cheese, yogurt, butter)

No beverages other than water

Here’s the quick list of what you can eat:

Vegetables including potatoes (but not corn as that’s a grain)

Fruits but in moderation

Seafood both fish and shellfish

Nuts and Seeds but not peanuts (legumes)

Coffee (but black; no sugar, no dairy)

Here is a good reference to Whole 30 .

The only sweet that I had on Whole 30 was Lara Bars (which are mostly made of cashews) and fruits / dates. The only drink I had other than water was LaCroix sparkling waters.

id=“zero-app”>Zero App

One of the more interesting things I did was use an app called the Zero app:

iOS

The idea of Zero app is that if you give your body more time to digest the food you consume then you will metabolize it better. I focused on reducing my eating window to lunch and an early dinner (i.e. 5 or 6). This reduced the window I ate in down to 17 to 18 hours. And while this is initially hard, after the first few days, it actually isn’t bad.

id=“exercise”>Exercise

I’ve never been a big exercise person but I did do 15 to 20 minutes on an elliptical 2 or 3 times per week.

id=“dining-out—just-say-no”>Dining Out - Just Say No!

I know that the U. S. has moved to a food culture that is largely based around dining outside the home / your personal kitchen. If you’re going to do Whole 30 then you pretty much can’t dine out at all. There is the rare occasion when you can find something simple like a piece of broiled fish or a steak but most restaurant dining is optimized towards flavor and that generally means copious amounts of butter / fat / sugar.

> *Note 2 : If you are going to weigh daily then you need to come up with a standard approach.*
>
> Scott Johnson (fuzzyblog)

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Week 7

Monica Dinculescu

I got deep into geocaching, friends. I found 3 caches in Arizona! I didn't sign any of them because I kept forgetting to bring a pen, so now nobody will know I was there.

Geocaching is mad in the US. We drove back from Arizona (flying? in this economy??), and there were HUNDREDS along the highway. First, who are these people that stop on the side of the highway to hide a cache? Second, who are these people who stop on the side of the highway to go look for a cache? Were you raised by wolves? Highways are dangerous yo, don't just stop on the shoulder unless your car is on fire.

I am filled with fury about cryptoart. FILLED. I follow someone on Instagram who has raised money for the national park service, and loves Joshua Tree, and a lot of her art is of nature, and she's been minting like 3 NFTs a week. Joanie Lemercier wrote how 10 seconds of cryptoart used more energy than their studio in 2 years. TWO YEARS. Imagine being an artist that is minting cryptoart in the same week that Texas residents are bankrupt because they couldn't pay their $17,000 heating bill in the winter because the Texas utilities companies are messed up and unregulated and can fuck with prices when supply is low. FILLED WITH FURY.

Related: I think part of the problem is that we're not making people build those absurd rigs to mine crypto anymore. In uni one of my friends built one of those babies to mine doge at home, and he could heat up his house in the winter from it! Now you just sign up for a service that mines in a data center and heats up the desert somewhere.

Also related: I 100% empathize with artists that selling art is hard and it's impossible to make a living, but as with everything, crypto isn't gonna solve anything. Unless you have a weird perpetuity clause in your NFT (is that a thing? I don't know. I make it a point to know as little as possible about crypto because I already have heartburn from regular life things and don't need more), the only people making money from cryptoart are the people who could already sell regular art for a shit ton of money. Otherwise nobody is going to pay any real money for your anonymous neon animated block, crypto or not.

I tweeted about this AND I now wrote about it and I am still filled with fury. Sorry to the 2 people who read these notes and also read this on Twitter. Nobody better email me about this; I cannot be convinced to not hate crypto, and I will just mute you with high prejudice and no guilt.

Honestly now that I am typing out these notes I'm also filled with fury about everything that's going on with Google and Research, so I don't think I've done anything other than be angry all week. lol what a time to be alive.

In the spare rage free seconds I've had left, I have been hard at work at getting meownica studio up and running. I have a special instagram account. I took the actual store out of maintenance mode. I enlisted my photographer friend Ashley (she is enormously talented, please hire her) to take photos so that I can update the store with real life product photos. This week is the week, I feel it.

Also canvas-sketch is dope and it does dope things (like use units that make sense)

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User-defined literals in Java?

Stephen Colebourne · Stephen Colebourne (Joda)

Java has a number of literals for creating values, but wouldn’t it be nice if we had more?

Current literals

These are some of the literals we can write in Java today:

integer - 123 , 12s , 1234L , 0xB8E817 , 077 , 0b1011_1010

floating point - 45.6f , 56.7d , 7.656e6

string - “Hello world”

char - ‘a’

boolean - true , false

null - null

Project Amber is also considering adding multi-line and/or raw string literals.

But there are many other data types that would benefit from literals, such as dates, regex and URIs.

User-defined literals

In my ideal future, I’d like to see Java extended to support some form of user-defined literals. This would allow the author of a class to provide a mechanism to convert a sequence of characters into an instance of that class. It may be clearer to see some examples using one possible syntax (using backticks):

Currency currency = `GBP`; LocalDate date = `2019-03-29`; Pattern pattern = `strata\.\w+`; URI uri = `https://blog.joda.org/`;

A number of semantic features would be required:

Type inference

Raw processing

Validated at compile-time

Type inference

Type inference is of course a key aspect of literals. It would have to work in a similar way to the existing literals, but with a tweak to handle the new var keyword. ie. these two would be equivalent:

LocalDate date = `2019-03-29`; var date = LocalDate`2019-03-29`;

The type inference would also work with methods (compile error if ambiguous):

boolean inferior = isShortMonth(`2019-04-12`);

public boolean isShortMonth(LocalDate date) { return date.lengthOfMonth()

Raw processing

Processing of the literal should not be limited by Java’s escape mechanisms. User-defined literals need access to the raw string. Note that this is especially useful for regex, but would also be useful for files on Windows:

/\/ user-defined literals var pattern = Pattern`strata\.\w+`; /\/ today var pattern = Pattern.compile(“strata\\.\\w+”);

Today, the `\` needs to be escaped, making the regex difficult to read.

Clearly, the problem with parsing raw literals is that there is no mechanism to escape. But the use cases for user-defined literals tend to have constrained formats, eg. a date doesn’t contain random characters. So, although there might be edge cases where this would be a problem, they would vert much be edge cases.

Validated at Compile-time

A key feature of literals is that they are validated at compile-time. You can’t use an integer literal to create an int if the value is larger than the maximum allowed integer (2^31).

User-defined literals also need to be parsed and validated at compile-time too. Thus this code would not compile:

LocalDate date = `2019-02-31`;

Most types which would benefit from literals only accept specific input formats, so being able to check this at compile time would be beneficial.

How would it be implemented?

I’m pretty confident that there are various ways it could be done. I’m not going to pick an approach, as ultimately those that control the JVM and language are better placed to decide. Clearly though, there is going to need to be some form of factory method on the user class that performs the parse, with that method invoked by the compiler. And ideally, the results of the parse would be stored in the constant pool rather than re-parsed at runtime.

What I would say is that user-defined literals would almost be a requirement for making value types usable, so something like this may be on the way anyway.

I like literals. And I would really like to be able to define my own!

Any thoughts?

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JPMS modules for library developers - negative benefits

Stephen Colebourne · Stephen Colebourne (Joda)

Java 9 introduced a major new feature - JPMS, the Java Platform Module System . After six months I've come to the conclusion that JPMS currently offers "negative benefits" to open source library developers . Read on to understand why.

Modules for library developers

Java 8 is probably the most successful Java release ever. It is widely used and widely liked. As such, almost all open source libraries run on Java 8 (as library authors want their code to be used!). Some libraries with a long history also still run on older versions. Joda-Convert has a Java 6 baseline, while Joda-Time has a Java 5 baseline. Others have a Java 8 baseline, such as ThreeTen-Extra.

Java 9 was released in September 2017, but it is not a release that will be supported for a number of years. Instead, it had a lifetime of six months and is now obsolete because Java 10 is out. And in six months time Java 11 will be out making Java 10 obsolete, and so on.

While most releases last six months, some are luckier. Java 11 will be a "long term support" (LTS) release with security and bug support for a few years (Java 8 is also an LTS release). Thus, even though Java 10 is out, Java 8 is still the sensible Java version for open source library developers to target right now because it is the current LTS release.

But what happens when Java 11 comes out? Since Java 8 will be unsupported relatively soon after Java 11 is released, you'd think that the sensible baseline would be 11. Unfortunately I believe many companies will be sticking with Java 8 for a long time. An aggressive open source project might move quickly to a Java 11 baseline, but doing so would be a risky strategy for adoption.

The module-path

Before discussing the JPMS options for open source library developers, it is important to cover the distinction between the class-path and the module-path. The class-path that we all know and love is still present in Java 9+, and it mostly works in the same way.

The module-path is new. When a jar file is on the module-path any module-info is used to apply the new stricter JPMS rules. For example, a public method is no longer callable unless it has been exported from the module it is contained in (and required by the caller's module).

The basic idea is simple, you put old fashioned non-modular jar files on the class-path, while you put modular jar files on the module-path. Nothing enforces this however, and it turns out this is a bit of a problem. There are thus four possibilities:

modular jar on the module-path

modular jar on the class-path

classic non-modular jar on the module-path

classic non-modular jar on the class-path

To be sure your library works correctly, you need to test it both on the class-path and on the module-path. For example, service loading is very different on the module-path compared to the class-path. And some resource lookup methods also work completely differently.

To complicate this further, JPMS has no explicit support for testing. In order to test a module on the module-path (which is a tightly locked down set of packages) you have to use the –patch-module command line flag. This flag effectively extends the module, adding the testing packages into the same module as the classes under test. (If you only test the public API, you can do this without using patch-module, but in Maven you'd need a whole new project and pom.xml to achieve that approach, so its likely to be rare.)

In the latest Maven surefire plugin (v2.21.0 and later) the patch-module flag is used, but if your module has optional dependencies, or you have additional testing dependencies, you may have to manually add them, see this issue and this issue .

Given all this, what should an open source library developer do?

Option 1, do nothing

In most cases, but not all, code that is compiled on Java 8 or earlier will run just fine on the class-path in Java 9+. So, you can do nothing and ignore JPMS.

The problem is that other projects will depend on your library. By not adopting JPMS at all, you block those projects from progressing in their modularization. (A project can only choose to fully modularize once all of its dependencies are modularized.)

Of course if your code doesn't run on Java 9+ because you've used sun.misc. Unsafe or something else you shouldn't have done then you've got other things to fix.

And don't forget that a user could put your jar file on the class-path or the module-path. Have you tested both? ie. The truth is that "do nothing" is not possible - at a minimum you have extra testing to do, even if it just to document that your project does't work on the module-path.

Option 2, add a module name

Java 9+ recognises a new entry in the MANIFEST. MF file. The Automatic-Module-Name entry allows a jar file to declare what name it will use if/when it is turned into a proper modular jar file. Here is how you can use Maven to add it:

org.apache.maven.plugins maven-jar-plugin

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(Data) Science or Witchcraft?

Target Brands, Inc · Target Tech

On my first encounter with it, around early 2010's, I was mystified. It sounded like witchcraft and I imagined the practitioners to be a coven of witches and wizards, all holding Ph. D.s in the dark art of "Data Science" and being respectfully addressed as "Data Scientists". It was believed they would magically transform haystacks into gold and then ask for your first-born in return as a reward for their service (a la Rumpelstiltskin) There is no denying the fact that the title "Data Scientist" is the most coveted one these days and has a nice ring to it. It's also true that data science has traditionally been a monopoly of mathematicians and statisticians. Obviously, developing statistical models and machine learning algorithms requires years of training and practice to specialize. In my opinion it is more of an art form driven by science and can easily be mistaken for magic.

It's common knowledge that the more experienced in life we get, the easier it is for us to make up our mind. For instance, "What diner to pick for a boy's-night-out?", "When to stay off highways to avoid being stuck in a traffic-jam?", "When to buy a house? When NOT to buy?", are all such decions we make everyday. This ability comes as result of years of learning from implicit experience (a.k.a unsupervised learning) and explicit instructions from parents, teachers, friends, family and media (a.k.a supervised learning.) Our brain builds models of the world, of the situations we have been in, of banal and extraordinary, of nice and not-so-nice, of appropriate and inappropriate etc. These models facilitate judgment, govern behavior and enable anticipation of likely outcomes. That's basically data science. The recent progress in large scale and high performance computing has opened doors for such complex calculations to be performed on-demand and much more efficiently than was possible before. Hence, the buzz!

At Target we operate in a guest-centric universe. We don't treat our guests as a statistic or as just a data-point in a trend. Our guests are setting their own micro-trends. Our focus is on carefully picking signals from our guests and learn what actually matters to them. Yes! We are seriously trying to understand each single guest's needs independently. Therefore, we strive for a fully personalized experience and not just making suggestions on what is popular out there. As John Fairchild has allegedly said:

id="style-is-an-expression-of-individualism-mixed-with-charisma-fashion-is-something-that-comes-after-style">"Style is an expression of individualism mixed with charisma. Fashion is something that comes after style."

At Target, Data Science and Engineering is a group that has gone beyond the conventional boundaries in terms of scale and applicability. To us the journey to the ultimate goal of 100% personalized experience began with the attention to detail, like what would our choice of algorithms be, how would we organize the data, what level of pre-processing would be enough, the frequency of compute cycles, strict SLAs on API exposed and so on. We were very clear that we needed to provide a consistent experience before worrying about fully personalized experience; establish reliability before making bunnies appear out of a hat. And our motto has been pretty simple, no matter what channel our guests take to interact with us we want their experience to remain consistent and pleasant.

What we do is driven by non-trivial problems that mandate an ensemble of approaches to be used. It is a slow and deliberate process of trial-and-error, experimentation with bleeding-edge algorithms/technologies and at times engaging in dialogue with peers and stake-holders, that can run for days. We have played with off-the-shelf algorithms as well as built new techniques. It is such a delight to witness the speed with which new ideas emerge from the insight built through models that use big data as input. We can understand our guests' behavior and needs better because of our omni-channel retail capabilities.

id="-it-is-such-a-delight-to-witness-the-speed-with-which-new-ideas-emerge-from-the-insight-built-through-models-that-use-big-data-as-input">" It is such a delight to witness the speed with which new ideas emerge from the insight built through models that use big data as input."

Data science and engineering teams that are tasked with solving business problems with quick turn-around times have a burning need to pre-process, ingest, store, process and retrieve large amounts of data fairly quickly and without compromising on security, quality or privacy. In the first year we have mostly focused on the design of long-running data pipelines, multi-phase compute workflows, highly scalable API layer and monitoring capabilities all aiming for the fastest turn around time. To name a few techniques, we have used Collaborative Filtering for behavior based recommendations, TF-IDF for feature selection, K-Means for clustering. Our exploration has a wider range though, as we are deeply interested in commoditizing data science for all our product teams within Target.

A key element in our ability to execute has been our choice to go with OpenSource almost every single time we had to make a technology decision with only a few exceptions here or there. We also rely on the contemporary Software Engineering and Management principles proposed by Agile development model, which has served us very well. Our engineers have a very wide variety of skills. Statistics, Machine Learning and Visualization techniques, delivered through Java, Scala, Python, Spark, R, Hadoop and many more technologies. That's how we transform abstract ideas into practical and well-engineered products.

We are doing some kick-ass engineering with the focus to build value-driven products through technology. And that's what data engineers do. For us, no idea is too big to try, and a failure is just a null-hypothesis we are trying to reject. Our dream is to go where no one has dared to go before and bring back the riches.

id="no-idea-is-too-big-to-try-and-failure-is-just-a-null-hy-

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standard-article(title: [dockerbash - Making docker exec -it Suck Less], author: [Scott Johnson (fuzzyblog)], source-name: [Scott Johnson (fuzzyblog)], images: (), paragraphs: ([Docker is a container technology that allows you to package up a series of different technologies under (generally) a *nix style operating system. As things deployed with Docker are generally deployed under a *nix style operating system, it isn’t uncommon to want to open a shell into your Docker environment for debugging purposes.], [You can easily do this with:], [The CONTAINER_HASH is a value like 311ab7fe0ea1. This value is fetched from a docker ps command like this:], [The term ‘police’ is just some bit of text that identifies the docker process that is running.], [Here’s an example of this output:], [What I’d really like is a command like this:], [and have that do the underlying work to generate the docker exec statement. A little bit of bash scripting gave me this script:], [class=“highlight”> #!/bin/bash if [-z $1] ; then echo “You need to specify the name of the container you want to get into like:” echo “dockerbash police” else pid = ` docker ps | grep $1 | awk ‘{print $1}’ ` docker exec -it $pid /bin/bash fi], [Save the lines above as dockerbash and make it executable. After that you can much more easily get a shell prompt inside your docker containers.], [id=“bash-references”>Bash References], [How to Set a Variable in Bash], [How to Use Awk to Get the First Variable], [id=“where-does-dockerbash-live”>Where Does dockerbash Live?], [You cannot store dockerbash in the project that you are deploying via Docker because it needs to exist on the machine that runs your Docker containers, not within your Docker container. My recommendation is that you have your DevOps tooling such as Ansible install this script.],), insert-map: (:), word-count: 321, edited-for-length: false, debug-mode: false,)

standard-article(title: [When Rails 7 Doesn’t Process application.js], author: [Scott Johnson (fuzzyblog)], source-name: [Scott Johnson (fuzzyblog)], images: (), paragraphs: ([This is more than a mildly embarrassing post to have to write but it is an interesting example of how habits can bite you hard – so, so hard.], [I recently wanted to start on an application and actually be current on everything: Rails, Bootstrap, etc. I have long admitted to having issues with respect to “modern” web development (specifically JavaScript / the asset pipeline) and I’m mildly starting to deal with those issues. I began with a BootrAils tutorial and started to work through it.], [I got all the way to the end and I noticed that I simply could NOT get the popovers to work. I don’t claim to have a great understanding of the asset pipeline but I do know that it is a preprocessor that should get invoked on every page request. I traced the problem back to application.js seemingly not getting processed (a view source didn’t show it). So, just to be sure, I added this to application],), insert-map: (:), word-count: 163, edited-for-length: false, debug-mode: false,)

You can easily do this with: class=“language-plaintext highlighter-rouge”> class=“highlight”> docker exec -it CONTAINER_HASH /bin/bash The CONTAINER_HASH is a value like 311ab7fe0ea1.

Scott Johnson (fuzzyblog)

standard-article(title: [Updating to the Latest Ansible on Ubuntu], author: [Scott Johnson (fuzzyblog)], source-name: [Scott Johnson (fuzzyblog)], images: (), paragraphs: ([One of the tricky bits about Ansible is that new features in the “language” always require the latest version of Ansible itself. And, while logical, this can easily bite you. Here’s an example:], [class=“highlight”> TASK [deploy_hate-language-modeling_systemd_start : just force systemd to re-execute itself (2.8 and above)] *** fatal : [aws_master2]: FAILED! => {“changed” : false , “msg” : “ Unsupported parameters for (systemd) module: daemon_reexec Supported parameters include: daemon_reload, enabled, masked, name, no_block, state, user” } to retry, use : –limit @/home/ubuntu/ansible/playbook_deploy_hate-language-modeling_master.retry], [What that translates to is that Ansible itself doesn’t know how to use the parameter daemon_reexec . And when you’ve been using an Ansible feature successfully and then it doesn’t work on the server, that usually is a rock solid signal “update ansible”.], [Here’s

standard-article(title: [Villa-Worthy Breakfasts We’d Actually Wake Up For], author: [Julia Youman], source-name: [Food52], images: (), paragraphs: ([If you’ve been around literally anyone in the past month, you’ve probably heard chatter about Love Island —whether it’s the UK version or the ever-iconic USA one. Maybe you’re all in. Maybe you’ve only absorbed it passively, against your will. Either way, it’s hard to escape.], [As someone who’d never tuned in before, I was surprised by the sudden cultural takeover. In the past few weeks alone, I’ve sat through full dinners where people debated their favorite couples, predicted drama, and ranked bombshells with passionate conviction. I finally had to get up to speed—if only to participate in what feels like a nationwide group project (slash networking event).], [Read More >>],), insert-map: (:), word-count: 111, edited-for-length: false, debug-mode: false,) how to do this on Ubuntu:],), insert-map: (:), word-count: 155, edited-for-length: false, debug-mode: false,)

standard-article(title: [What BigCommerce Has Been Doing with AI], author: [Mahendra Kumar], source-name: [BigCommerce Engineering], images: (), paragraphs: ([In June, we shared our winning Hackathon project that combined three AI tools into one package.], [Related Video: AI Considerations & Lessons We've Learned], [The work didn't stop there.], [Since then (and before that time), we've been developing some really amazing features that help elevate BigCommerce's merchant experiences, and we're looking forward to releasing this application in the near future.], [In this chat with Heather Barr, Developer Community Program Manager, we'll show you some demos, deep dive into the code, and explore some architectural diagrams and data flows.], [id="heather-barr-developer-community-program-manager">Heather Barr, Developer Community Program Manager], [Background : I've been on the Developer Relations team at BigCommerce since 2020, focusing on our Developer Community . Since 2018, I've been part of the tech space with experiences ranging from software engineering to sales and support. I work alongside the Developer Advocates here at BigCommerce and am always looking to connect with developers who build on our platform.], [Fun fact : I live in a van down by the river! Just kidding! But I do travel the US and camp in my self-converted camper van a few times a year. ⊠], [Connect with Heather on X .], [id="mahendra-kumar-vp-of-data-and-software-engineering">Mahendra Kumar, VP of Data and Software Engineering], [Background : I've been at BigCommerce for six years. My primary area focuses are data engineering, machine learning, and search technology. My hobbies include hiking and playing tennis. I also enjoy volunteering for community projects.], [Fun fact : In my twenties, I was able to speak seven languages. Although, now I am proficient in three languages only.], [Related Article: Use of Kafka and Kafka Streams at BigCommerce], [id="krishna-teja-are-senior-data-engineer">Krishna Teja Are, Senior Data Engineer], [Background : I have six years of a software background and have been at BigCommerce for the past two years. I am a driven individual who is passionate about my work.], [Related Demo: Watch Krishna's Walkthrough of His Hackathon Project], [id="eugene-kuzmenko-software-engineer-team-lead">Eugene Kuzmenko, Software Engineer Team Lead], [Background : I have been at BigCommerce for almost four years. This company has an amazing culture, and I like working on the Data Team because we do really good stuff. I've been in software development for almost ten years and enjoy winter sports like snowboarding.], [Open Roles: Explore Your Next Engineering Opportunity], [Fun fact : I don't like blue jeans, and I don't have them in my closet.], [We're excited to give you a peek behind the curtain with this technical session. Like Mahendra says in this video, "Fasten your seatbelts!"], [style="overflow: hidden; width: 100%; padding-top: 62.5%;">], [id="additional-resources">Additional Resources], [BigCommerce & Google's BigAI Hackathon: Results & Recap], [BigCommerce X Google BigAI Hackathon YouTube Playlist], [BigCommerce Developers YouTube Channel], [BigCommerce Developer Community], [BigCommerce Developer Blog],),

standard-article(title: [Registering a Domain with Name Silo], author: [Scott Johnson (fuzzyblog)], source-name: [Scott Johnson (fuzzyblog)], images: (), paragraphs: ([So I needed to register a . IO domain today. Amazon had .io domains at $71 per and Google had them at $60 . A quick check with my guru of cheapness , Nick Janetakis, said:], [NameSilo.com], [And he convinced me. Here are some tips:], [Do not accept any of their options.], [Just the domain, Sir, just the domain.], [Use DDOS protection from CloudFlare instead of their option. CloudFlare is free.], [Once you are registered then the option you likely need is to create an A record. This is done by selecting the domain from the Domain Manager view and then clicking the blue icon.], [Amazon is way prettier but double the price is a hard sell.], [After you create it, wait some period of time and ping it. Verify the ip address is yours.], [id="references">References:], [Review of NameSilo], [Mass Migration to NameSilo],), insert-map: (:), word-count: 142, edited-for-length: false, debug-mode: false,) insert-map: (:), word-count: 469, edited-for-length: false, debug-mode: false,)

I also enjoy volunteering for community projects.

Mahendra Kumar

standard-article(title: [Adding an Includes Clause to ActiveRecord and Watching the Joy Flow], author: [Scott Johnson (fuzzyblog)], source-name: [Scott Johnson (fuzzyblog)], images: (), paragraphs: ([I've written in the past about watching your SQL queries stream by in the Rails console and how seeing, well, stupidity / things that look wrong can help guide you to things you need to find. Here's an example I witnessed recently:], [Metric Load (2.5ms) SELECT metrics.date_created_at, metrics.int_val, metrics.float_val, metrics.metric_type_id FROM metrics WHERE metrics.habit_id = 2 AND (date_created_at >= '2019-11-01') AND (date_created_at], [class="highlight"> def total_this_month today = Date . current date_start = DateCommon . first_date_of_month (today) date_end = DateCommon . last_date_of_month (date_start) total_from_date_to_date (date_start , date_end) end], [And here's the total_from_date_to_date method:], [class="highlight"> def total_from_date_to_date (date_start , date_end) self . metrics . select (Metric :: TOTAL_FIELDS). where (["date_created_at >= ?" , date_start]). where (["date_created_at <= ?" , date_end]). includes (:metric_type). map (& :amount). compact . sum end], [Just adding a simple .includes(:metric_type) clause to the where statement fixes this and makes that data available to the underlying .compact.sum operation. And, yes, to fix a performance problem this easily really does make the joy flow for a developer.],), insert-map: (:), word-count: 224, edited-for-length: false, debug-mode: false,)

standard-article(title: [Rails 7 Madness: No such middleware to insert before: ActionDispatch::Static], author: [Scott Johnson (fuzzyblog)], source-name: [Scott Johnson (fuzzyblog)], images: (), paragraphs: ([In tonight's category of "Crazy Rails Errors that Even I haven't hit", I was deploying a new app via HatchBox and I hit this one:], [When I dug into the underlying failure there was this bit of code:], [def assert_index (index , where) i = index . is_a? (Integer) ? index : index_of (index) raise "No such middleware to insert #{ where } : #{ index . inspect } " unless i i end], [I did the obligatory googling only to find nothing in the past year and absolutely nothing related to Rails 7.], [I was able to replicate this by logging in with either:], [or], [After stumbling around a bit – this was post midnight late night code fu – I thoguht to myself:], [What if the issue is that it is looking for middleware and it can't find any?], [One of the previous googles from 2015 or so said that if you have this issue then remove the font_assets gem (I didn't have it but omitted that part of the research above). What I did was look for asset pipeline middleware and I found the smart_assets gem which I dropped into Gemfile:], [A quick bundle install and I ended up with a failing deploy ... but a different failure:], [Visit the Troubleshooting guide for more information: https://vite-ruby.netlify.app/guide/troubleshooting.html#troubleshooting], [And that, in modern computing, alas, is progress .],), insert-map: (:), word-count: 287, edited-for-length: false, debug-mode: false,)

standard-article(title: [Essential Technology for a Team Offsite], author: [Scott Johnson (fuzzyblog)], source-name: [Scott Johnson (fuzzyblog)], images: (), paragraphs: ([I'm in the process of wrapping up a team offsite even though I was ostensibly the lead technologist present, I was not actually prepared in terms of bringing the right technology with me to facilitate the team's needs. The single worst omission was the lack of an HDMI cable without which, I saw three tech folk (me, other engineer #1, team coordinator) fail to be able to connect a Mac, an iPad and a Windows box to a Chrome Cast equipped TV.], [Here is my list of the technical stuff I will never again be without for a team meeting:], [A couple of USB sticks. Even today in the age of Google Drive and DropBox, USB sticks can still be more convenient], [An Anker USB battery just in case], [2 lightning cables, at least one of which is 6 foot like this Amazon cable], [1 6′ USB type C cable], [1 USB 2 cable], [1 power strip with USB ports], [1 extension cord with USB ports], [1 Verizon Mifi Hotspot able to share connectivity if WiFi fails], [1 ChromeCast], [1 6′ HDMI Cable], [1 USB Speaker like a Beats Pill], [This is roughly enough technology to support a team offsite of 6 to 8 people (imho).], [Note : I use a MacBook 2015 classic so I still have real ports. If you are using newer MacBooks then you likely need dongles as well.],), insert-map: (:), word-count: 234, edited-for-length: false, debug-mode: false,)

standard-article(title: [Python Machine Learning Best Practice - Lock Down Your Versions], author: [Scott Johnson (fuzzyblog)], source-name: [Scott Johnson (fuzzyblog)], images: (), paragraphs: ([I recently took a Python and Ruby (but mostly Python) Machine Learning data pipeline live and, almost immediately, hit a pretty significant problem. The issue was this line of code:], [This, which worked, at least for a while, perfectly on my development box. And then I deployed the code and that caused my requirements.txt file to be fulfilled and that's when the trouble started. As I hope you know, requirements.txt is a list of any software components that your project uses along with, sometimes, the version numbers of those software components.], [Now I mostly work over in the happy go lucky land of Ruby and the Ruby world has a pretty mature ecosystem at this point where components change but not terribly frequently. The Ruby equivalent to Python's requirements.txt is called Gemfile and like requirements.txt, a Gemfile can include the version of a component. But, because the Ruby world's components tend to be mature, I haven't worried about locking down version numbers for ages. Yes, yes, I know that this is bad but I'm human and fallible.], [So what happened when my code was deployed is that this command was executed:], [as part of the underlying Docker build process. And this brought in whatever the latest version of our underlying Machine Learning libraries happened to be. And that broke the line of code above which then led to the expected scramble to:], [figure out what version I had on my development machine], [update requirements.txt], [shut down the cluster of boxes running our pipeline], [redeploy the code], [Here's the sample recommendation:], [Always, immediately , lock down the version of any machine learning libraries that you put into requirements.txt. These libraries are changing regularly and their changes can very easily break your code. You need to do this when you add one of these libraries to requirements.txt.],), insert-map: (:), word-count: 336, edited-for-length: false, debug-mode: false,)

standard-article(title: [I am not a software engineer], author: [0xADADA], source-name: [0xADADA], images: (), paragraphs: ([Software engineering, knowledge work, or most broadly any type of white collar labor is, primarily, the production of our own psychology. Our mental capacity is our means of production. We embody the values of our work. Our identity is tightly bound up with our careers, knowledge, skills, methods, company, and our industry— we are our work .], [Having our identities bound to our work, and most dangerously, to an employer severely limits ones ability to negotiate wages when that identity is immediately laid bare in monetary value— our human value is our wage .], [Beyond wages, this mindset limits the horizons of what is possible to do with our own lived time.], ["I need to do software engineering forever, for I am a software engineer". We can create art with software, we can also create software as art, we can also choose to not make software, and we are still the same person. Software is something we create; we also create ourselves.], [As someone who has spent a lifetime in software engineering, who has credentials in the field, and who's hobbies and interests have, at various times, fallen into the bin of "software engineering", I have often identified my worth with my current job, wage, and employer. I have put all my eggs in one basket. Identifying who we are with what we do is a cognitive limit.], [This is to say, I refuse to self-identify only as invested in my work, my career, my achievements. This might have been okay where we talked about the idea of a career as a vocation, but not now—when the only horizon of intelligibility available according to the dominant mindset is one where we understand our selves only in terms of pure utility, or economic productiveness. This I refuse.], [I do software engineering. I am not a software engineer.], [Travel Writing - Redux, Italy & Switzerland Edition], [I am not a software developer], [The Disappearance of Lived Time],), insert-map: (:), word-count: 327, edited-for-length: false, debug-mode: false,)

standard-article(title: [If You Pair Program Buy Tuple Now], author: [Scott Johnson (fuzzyblog)], source-name: [Scott Johnson (fuzzyblog)], images: (), paragraphs: ([As I've written in the past, I am an unabashed fan of pair programming. Pair programming is one of the few techniques that I can credibly point to that really, really, really improves software quality. And since I'm a remote developer, I really, really need good pairing tools. The unfortunate reality of pair programming is that there really aren't good tools for pairing – until Tuple.], [The current suite of tools for pairing include things like:], [Google Hangouts / Google Meet - slow, craptastic user interface], [Skype - don't get me started; Microsoft gutted Skype and left it for dead in a alley], [Slack - only available on paid slack; slow, uses a lot of CPU], [While there have been good tools for pairing in the past, notably ScreenHero and Apple Remote Desktop, none of these tools are available today. ScreenHero got bought by Slack and "integrated" while Apple Remote Desktop, well, I seem to have been the only user who remembers it.], [The current tool I'm using is Tuple and it is simply outstanding . Here are some of the reasons:], [It just works. For pairing to be something that you just do on an ad hoc basis, the tool needs to be utterly unobtrusive. Tuple is brilliantly in its minimalism.], [It doesn't slow everything on the machine to a crawl.], [The audio quality is fantastic.], [It is fast.], [Here are the disadvantages:], [It is expensive; the model is a monthly fee.], [It is OSX only.],), insert-map: (:), word-count: 248, edited-for-length: false, debug-mode: false,)

standard-article(title: [Hackathon Winners Part 2: AI Simplifies Product Descriptions], author: [Qifeng Zhao], source-name: [BigCommerce Engineering], images: (), paragraphs: ([Greetings, all. I'm Vincent Zhao, a Software Engineering Manager in BigCommerce's Sydney, Australia office.], [Related Articles: Meet More Hackathon Winners], [Recently, my ChatGPT project placed second in our bi-annual Hackathon. With so much focus on AI, you probably won't be surprised to learn that the Best Project Award went to another AI-focused project.], [I'm pleased to introduce

standard-article(title: [Warai Otoko (笑い男) Animated Logo], author: [0xADADA], source-name: [0xADADA], images: (), paragraphs: ([The Laughing Man 笑(わらい) い 男(おとこ) (warai otoko) is a fictional character in the anime series Ghost in the Shell: Stand Alone Complex .], [This lil' project is an animated SVG using CSS transforms to rotate the text.], [The Laughing Man logo is an animated image of a smiling figure wearing a cap, with circling text quoting a line from Salinger's novel The Catcher in the Rye , which reads:], ["I thought what I'd do was, I'd pretend I was one of you to our winner, Luke Eller, a Staff Software Engineer. His project leverages the power of AI to make creating product descriptions super simple.], [I interviewed Luke about what inspired his project, challenges he faced during the Hackathon, and some things to keep in mind when you work with AI. Before you watch our chat , here are some of the guidelines we have in place for our Hackathons:], [This Hackathon ran for two consecutive workdays.], [There is one prize category with first and second place (that's me) prizes awarded.], [After every team demos their projects, voting opens for all Sydney-based team members. The team who receives the most votes is the first place winner.], [style="overflow: hidden; width: 100%; padding-top: 62.5%;">], [Thanks for taking the time to watch our discussion . I hope you enjoy it!], [If you have any questions for Luke, you can email or message him on LinkedIn.],), insert-map: (:), word-count: 223, edited-for-length: false, debug-mode: false,)

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ANALYSIS

those deaf-mutes."], [This character deeply resonates with me; he/she is a corporate hacktivist infiltrating micromachine manufacturing corporations that hide an inexpensive cure to a debilitating disease in order to profit from their more expensive but proprietary micromachine treatment. They are able to hide their physical presence by editing themselves out of video feeds and surveillance systems in real-time by by superimposing the animated logo over their face.], [The Laughing Man logo has been co-opted by pop culture in advocacy for the Electronic Frontier Foundation and the loose hacktivist collective Anonymous (using the latter's motto We are Anonymous. We are Legion. We do not forgive. We do not forget. Expect us.], [Code is up on github],), insert-map: (:), word-count: 197, edited-for-length: false, debug-mode: false,)

IN BRIEF

Tom Breihan, Stereogum: The Brooklyn drill rapper Pop Smoke's career was just getting started when he was shot dead in a Los Angeles home-invasion robbery in 2020. When Pop died, he was just 20 years old. In 2025, one of his attackers was sentenced to 29 years in prison. Since Pop's passing, his estate has released two posthumous albums , the first of which went double platinum. Now, Pop Smoke's brother is using his name to start a coffee shop, and he needs some experienced baristas.

The post Experienced Baristas Needed To Honor Pop Smoke's Legacy appeared first on Stereogum .

Carlos Becker, Carlos Becker: I wanted to set up a high available nats-streaming-server cluster, but couldn't find a "quick" guide on how to do it.

Carlos Becker, Carlos Becker: I've been wanting to write this for a long time, just to clarify my thoughts on the subject. Now, on vacations, I took a couple of days and finally did it. This is a personal opinion based on my personal experience and tons of books I have read, and I am not, by any means, the supreme holder of the truth, so you will probably disagree with me at some point.

Carlos Becker, Carlos Becker: I got a MacBook Pro 14″ with an Apple M1 Pro SoC, 16GB of memory and 500GB of disk a couple of weeks ago, and wanted to write my impressions about it, since a lot of people ask.

Carlos Becker, Carlos Becker: This past weekend I decided I need to clean up my GitHub profile. In this post I'll write about why I cleaned everything up and also how I did, as well as some initial results.

Scott Johnson (fuzzyblog), Scott Johnson (fuzzyblog): class="center">

Tailwind is a CSS grid library that is an alternative to Bootstrap and, well, something I'm increasingly using. Here are some handy, dandy Tailwind tools:

Tailwind Home Page

Tailwind Tutorial

Tailwind Button Generator (written in Elm !!!!)

Tailwind Grid Generator

Tailwind Toolbox

Tailwind Builder

0xADADA, 0xADADA: From comic book stores to Facebook to message boards, it seems that the "Keep Calm and Carry On" poster is living a new life as pop-culture cool while people remix its' message with both quirky or banal messages.

This attempt tries to speak most directly to its underlying message of consumerism, amidst the current climate of the Occupy Movement.

Carlos Becker, Carlos Becker: I've been working on GoReleaser for more than a year now, and one of the things that was bothering me the most was fpm .

Scott Johnson (fuzzyblog), Scott Johnson (fuzzyblog): So I just dropped a pill on the ground as I took my daily vitamins and I picked it it up and said "5 second rule" before popping it in my mouth. And then it struck me:

In the age of covid, the 5 second rule no longer applies. Covid can live on surfaces for up to 3 days. MIT Technology Review

LASZLO GYORI, SOLUTION ARCHITECT @ TOPTAL, Toptal Engineering: Trying to decide between gRPC and REST? This guide goes beyond surface comparisons to focus on the performance and operational trade-offs that determine which API framework is right for your next build.

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Wide Herald

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